Alexandre Dalban

Citizenship: USA, EU | **Settled Status:** UK

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Education

CentraleSupélec, *MSc in Artificial Intelligence (M2)* — Paris, FR | Sept 2025 – Nov 2026

Boston College, *BS in Mathematics and Computer Science* — Boston, USA | Aug 2021 – May 2025

Experience

Applied AI Research Intern, *Siemens AG* — Munich, GER | May 2026 – Nov 2026

- Automating a manual step in manufacturing (CAD to CAM): built a model that automatically detects machinable features (holes, pockets, slots, etc.) on 3D part designs, replacing work currently done by hand. Uses self-supervised learning on sampled point clouds, building on Meta's Sonata and Xiaomi's Utonia.
- Experience using graph neural networks for classification.
- Current approach achieves panoptic quality (95%) on par with state-of-the-art on the MFInstSeg benchmark.

Quantitative Finance Intern, *LeCap Asset Management (Hedge Fund)* — London, UK | Summers 2024, 2025

- Researched and validated 100+ alpha signals for systematic equity portfolio strategy, supporting portfolio construction.
- Built production Python tooling for the live trading environment, including an automated weekly reporting system evaluating 150+ signals across 8 performance metrics, replacing manual analysis.
- Built supervised forecasting pipeline spanning classical ML through transformer architectures, training on 90 proprietary indicators and Axioma factor-residualized returns across the top 1000 North American equities; best model achieved 54% daily directional accuracy vs. ∼50% coin-flip baseline.
- Developed PPO reinforcement learning agents with reward based on rolling 5/10/20-day realized Sharpe and P&L, with custom cross-sectional and cross-temporal features (rolling autocorrelation, recent relative performance); agents output dynamic indicator weightings producing a composite prediction signal.

Projects & Academic Research

Domain-Specific LLM for Reliability Engineering [*GitHub*] — CentraleSupélec

Supervised by Dr. Zhiguo Zeng (Industrial Engineering Lab)

- Built the first large training set for reliability engineering (failure analysis, MTTF, Markov chains, fault trees, etc.) by bootstrapping 56 textbook seeds into 866 verified Q&A pairs via a cross-model Self-Instruct pipeline, replacing the classical Self-Instruct same-model consistency filter, which we showed is trivially satisfied in technical domains.
- Fine-tuned Qwen3-8B with QLoRA via the Unsloth framework, then applied GRPO with DAPO-style dynamic sampling for reasoning refinement on small data; ran 26 ablation experiments on the HPC cluster (A100 GPUs, SLURM).
- Achieved +18.6 pp accuracy (63.6% → 82.2%, $p = 0.002$, 10-fold cross-validation with Wilcoxon signed-rank test), confirming genuine domain-knowledge acquisition rather than noise; paper in preparation for 2027 conference submission.

Phantom: Cryptocurrency Return Distributions Forecaster [*GitHub*] — CentraleSupélec

- Built a Transformer with a 2-layer cross-attention decoder jointly predicting Student-t return distributions across 1–30 day horizons for 362 cryptocurrencies using 120 day lookback.
- Two-stage pipeline: pretrained on 413 diverse assets (crypto, equities, FX, commodities) using identical OHLCV features, then fine-tuned on the 362-crypto universe.
- Out-of-sample (Jan 2025–Mar 2026): Rank IC = 0.124 at the 10-day horizon.
- Ran 8-version ablation isolating each design choice: synthetic SDE pretraining transferred zero real-world signal, multi-asset pretraining alone was directionally flat, expanding to 362 cryptos improved long-short diversification, and 4-hour bars degraded results due to sample overlap; self-audited the evaluation, correcting a $\sqrt{10}$ Sharpe-inflation error and publishing survivorship caveats.

Technical Skills

Languages: Programming: Python (advanced), Java | **Human:** Native French and English
Deep Learning: PyTorch, TensorFlow, Hugging Face, CUDA, Sentence-Transformers, Cross-Encoders
Architectures: Transformers/attention, VAEs, self-supervised & generative models, time-series forecasting
Quantitative Finance: Alpha research, factor models (Axioma residualization), systematic trading, signal evaluation, long-short strategies, return distribution forecasting (Student-t)
Data Science: Pandas, NumPy, Scikit-learn, Apache Spark, SciPy, FAISS, Matplotlib
LLM Training & RL: QLoRA, GRPO, PPO, Unsloth, synthetic data generation (Self-Instruct)
Tools: Git, asyncio/aiohttp, WebSocket, REST APIs, Docker, SQLite, LaTeX, SSH, SLURM, Bash, Claude Code
Achievements: 1st in CentraleSupélec GreenAI hackathon, 2nd in HackEurope Solana prize ($1000), 13th in France at IMC Prosperity 4 (458th out of 18,800+ teams globally)